

Virtual Model Control for programmable passivity in compliant hand

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Abstract.

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Introduction

Biological manipulation achieves remarkable adaptability through dynamic modulation of mechanical impedance, being able to respond to forces differently according to the task [1]. Humans continuously vary finger stiffness via muscle co-contraction [2, 3], transitioning between compliant exploration and rigid grasping with identical musculoskeletal structure and adapting the force response to compensate for instabilities [4]. This capacity for programmable passivity distinguishes biological from robotic manipulation [5].

Compliant robotic hands embody many of these biological advantages by design [6], with different solutions exhibiting various stiffness properties over some orders of magnitude and inherent robustness [7]. Passive mechanical complexity can generate adaptability and diverse intelligent responses through environment-dependent interactions [8], and passivity has been shown to enable behavioral diversity even in systems with minimal active control [9]. The trend in robotic hand design has accordingly shifted toward more compliant architectures, which allow them to adapt their shape to objects, simplify control, tolerate impacts, and enable safe human interactions [10].

Although compliant, tendon-driven hands offer stability and robustness to perturbation by construction, these properties are not tunable and limit manipulation to specific response behaviors. Building controllers on top of these capabilities will enhance the system while simplifying the force distribution problem in manipulation [11]. The resulting architecture introduces an active compliance on top of the passive one. By superimposing two dissipative systems, the result is stable by construction [12], and the physical compliance provides a safe fallback. This also allows bandwidth separation: the body handles high-frequency disturbances through mechanical filtering, while the controller addresses lower-frequency objectives such as stiffness shaping and force tracking, mirroring the biological hierarchy in which intrinsic musculoten-

don impedance provides immediate stability, while spinal reflexes and cortical planning handle progressively slower adaptations [13, 14]. Model dependency is also reduced, as the active compliance only acts on the kinematics without canceling the dynamics [15].

Various strategies have attempted to replicate human capabilities on grasping and expressive tasks [16]. Although effective methods exist for tactile interaction and modulation, most rely on learning or vision-based approaches [17] and lack compliance-level intuitive modulation and integration with passive hardware [18]. Standard systems often rely on position control [19], which can be unreliable for contact-rich manipulation because it ignores task dynamics and leads to rigid interactions. Fixed compliance alleviates this rigidity but constrains the hand to a single response behavior, with no means to adapt across tasks.

Variable stiffness actuators attempt to dynamically modulate the force-displacement relationship by tuning mechanical properties [20, 21, 22]. Previous works have introduced them in manipulation through shape-memory materials [23, 24] or physical springs placement [25]. However, none provide an effective solution for fast, dynamic, control-wise stiffness modulation, and all constitute a burden in terms of hardware complexity.

Stiffness modulation through control becomes the viable solution for scalable, intuitive algorithms. To avoid unreasonably high stiffness and alleviate the burden of modeling every interaction [26], intrinsic compliance should be incorporated into the control loop [27] through impedance control [28]. Standard impedance control seeks to cancel the natural dynamics to modulate end-effector behavior in terms of apparent stiffness, damping, and inertia [29], allowing the motion-force relationship to be independently tuned in a passive framework [30]. Effective methods have been developed for human-like adaptation of force and impedance [31], and pure impedance control has been applied to dexterous hands [32, 33], including combinations with passive joint compliance for robustness against load peaks [34].

However, impedance control is fundamentally model-dependent, requiring precise knowl-

edge of robot dynamics. Such accuracy is rarely attainable in complex hardware like tendon-driven compliant systems. At a more fundamental level, not all impedance control algorithms are inherently passive [35], and determining the optimal task-impedance pair remains a significant challenge for multi-fingered hands. Virtual Model Control (VMC) shows promise in overcoming these limitations by building on top of a compliant system, giving the robot *hints* rather than commands [36] through the superposition of a passive virtual system that preserves the dissipative properties [37].

Motivated by the dexterity of the human hand [38], we target tendon-driven compliant hands as the platform for passive and tunable virtual controllers. The anthropomorphic ADAPT Hand [39], here with modular fingers from a benchmarked design [40], replicates this morphology with biomimetic compliance distributed across tendons, finger springs, and skin [41]. We augment it with a passive VMC architecture that overlays virtual compliance on top of this mechanical baseline (Fig.1), enabling tunable modulation of the existing physical response.

Virtual Model Control tunes the passive stiffness while guaranteeing resilience and robustness to real-world uncertainty and object properties, preserving the stability of passive systems [42, 43]. It addresses the inherent complexity of tendon-driven systems by mapping virtual task-space forces to joint torques via virtual work [44], bypassing the need for a high-fidelity dynamic model. Rather than modulating apparent impedance parameters [45], VMC introduces virtual mechanical elements whose interactions are computed by the Principle of Virtual Work [46]. The approach is passive by construction and applies to all systems, including underactuated ones. Control parameters become physical properties of the virtual structure [47], enabling real-time stiffness variation [48], proprioceptive force sensing, force and stiffness tracking, and simulation of muscular co-contraction. This allows to operate in the full force-displacement space: stiffness defines the slope of the relationship, while force sets the operating point. Tuning both is needed for robust manipulation, as it leads to feedforward force

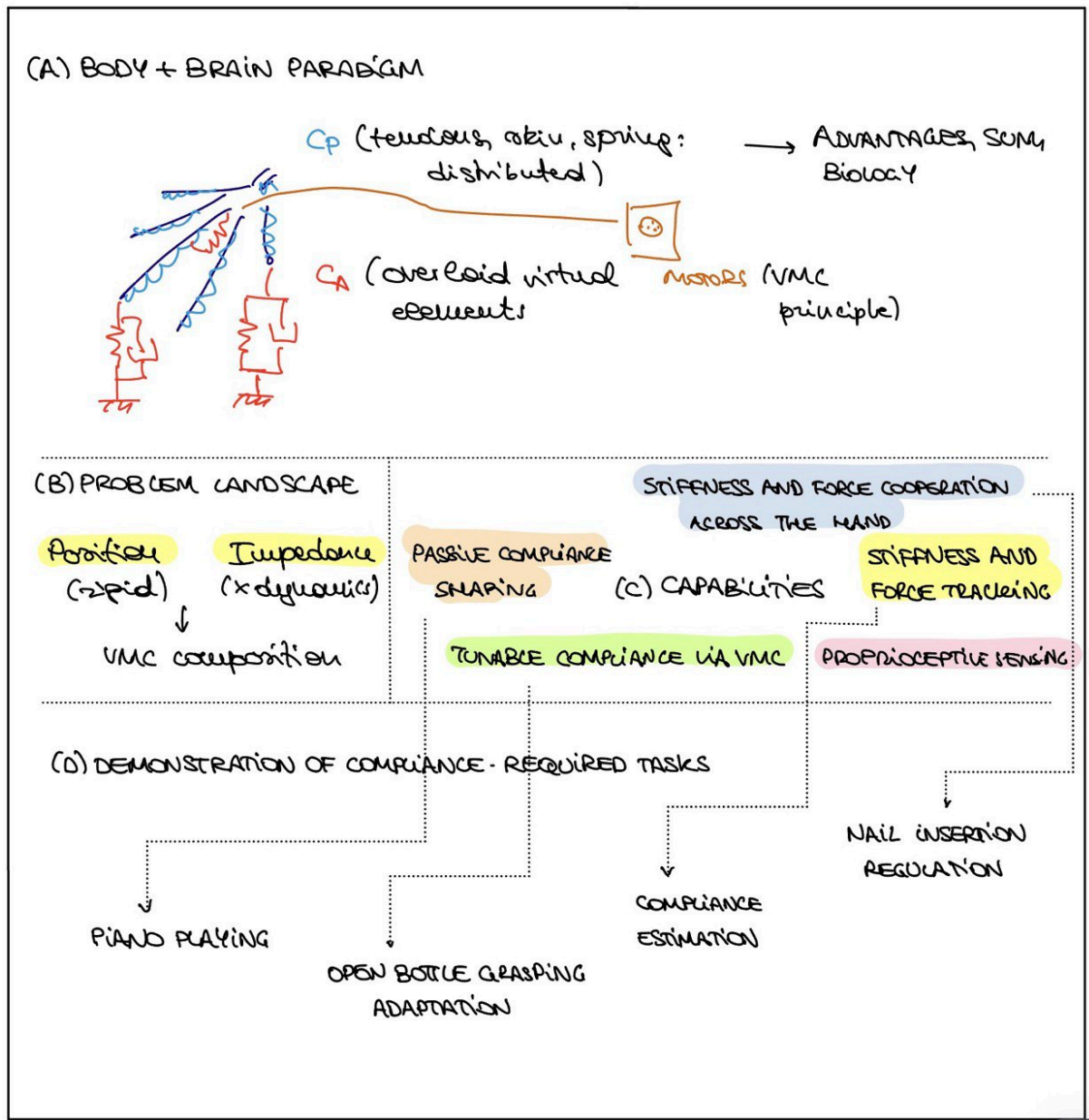


Figure 1: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth. **Note:** in (B) Problem landscape, also add VMC allows to do... and then the force/displacement graph.

control with disturbance rejection via stiffness.

While prior VMC work [46, 47] established compliant reaching and passivity-based tuning, it has not been applied to multi-fingered tendon-driven hands, nor extended to the online stiff-

ness modulation and proprioceptive sensing used here for dexterous manipulation. We present a kinematic-only VMC framework for tendon-driven compliant hands that allows variable passivity and online stiffness modulation. By combining a tunable passive virtual system with physical compliance, the hand enables intuitive definition of virtual elements, custom stiffness shaping, proprioceptive force and stiffness estimation, and independent control of force and stiffness. Using virtual springs and dampers, we achieve tasks ranging from guitar playing in different stiffness regimes and stiffness adaptation during dynamic grasps, to classification of object compliance and force control during object picking, all from a single hardware platform without mechanical reconfiguration.

In this study, we first validate that the virtual layer reproduces and shapes the passive response of the compliant fingers through contact, illustrated by guitar playing across stiffness regimes and by sustaining weights under various stiffness. Then, tunable compliance is analyzed with respect to the emerging grasp types during in-hand manipulation and dynamic grasping. We show that the knowledge of kinematics and virtual stiffness allows proprioceptive estimation of contact force and classification of object stiffness. Finally, the model-based formulation allows independent optimization of force, position, or stiffness with improved tracking performance, demonstrated by adapting grip force while grasping and picking an apple.

Results

Passive compliance shaping

Why does it improve manipulation? Passivity theorem, adaptability to the environment and invariance to noise, sensing errors, and object properties. We are robust to unplanned contacts, and while a rigid hand is regardless of what it touches, introducing compliance makes the hand pose depending on it and absorbing interaction energy. Compliance becomes a manipulation decision that exists before any control. Compliance also acts as bidirectional filter: outward,

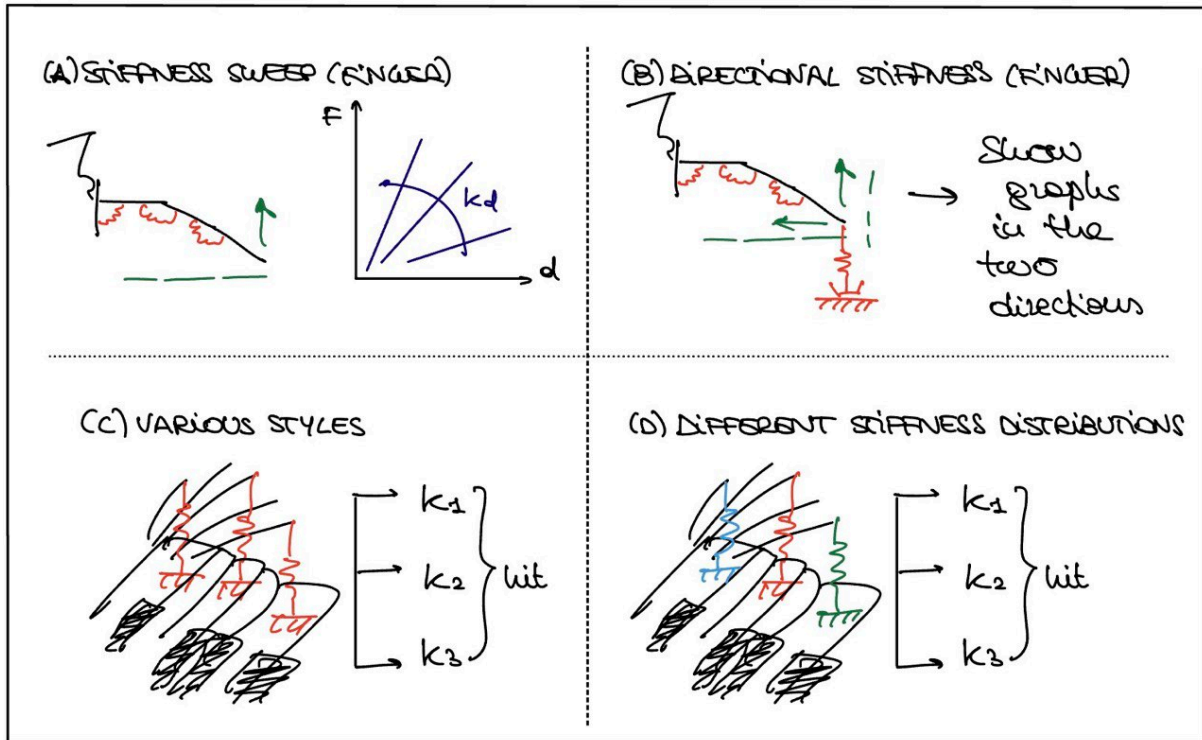


Figure 2: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

it absorbs dynamics and noise before reaching the contact; inward, it smoothens deformations, providing adaptability.

Notes: *extend to damping and combine analysis of damping coefficients with dynamic piano playing, maybe always using MIDI outputs.*

Tunable compliance via VMC

Why does it improve manipulation? Fixed compliance, in any case, defines fixed interaction behavior. But different phases in reality require different stiffness profiles (soft approach, firm hold, and then gentle release), across objects (egg vs hammer), and even across fingers simultaneously. VMC allows this in a stable way and without hardware changes, with the controller not *fighting* the body yet *modulating* it. Soft-to-rigid spectrum can be obtained on-demand and

model-predictably.

Notes: *adapting stiffness online, loose stability. We have passivity modulation even at the loss of passivity (see theoretical guarantees). Passivity can be observed as long as the hand is moved carefully/slow enough. In (B), we are online adding (not just modifying) everything flexibly for different compliant response. Gaussian springs also allow a different profile, softer when closer and harder elsewhere.*

Gaussian: *Modulate between the states of the system, modulating continuously through that. The rest is calibration. Stiffness also depends on the displacement. Slope will be modulated according to d , so we can modulate the stiffness. This is compatible with passivity.*

Proprioceptive sensing

Why does it improve manipulation? External force sensors add cost, fragility, and wiring. Compliant hardware unifies both (coarse) sensing and safety: the deformation that absorbs impacts is the same that encodes contact force. This overcomes position-based control in allowing compliant force estimation. *Transmission gain* take into account tendon losses, unmodeled dynamics, and body compliance.

Notes: *explore a bit. Tune better the compliance to match the internal spring to the external one. If we have larger displacement, we will have larger sensitivity (think about it). We want to understand sensitivity. Important thing: why we don't have monotonicity? It is the saturation of the force (it happens at more or less 120 N/m, and we are at 200), so it is very very easy to motivate. But put this.*

Stiffness and force tracking

Why does it improve manipulation? Stiffness alone defines disturbance rejection (the slope), but not the operating point; force sets the operating point but not the response to perturbations.

Manipulation requires both, for having a target contact force but also a stiffness that prevents slip. The optimization pathways (stiffness or reference for force) give closed-loop control over the full force-displacement relationship.

Notes: *put (C) Exploitation of "null space" with showing the graphs F , d , K , using the relationships to really show what happens into the system.*

Stiffness and force cooperation across the hand

Why does it improve manipulation? As humans independently regulate grip force and grip stiffness via co-contraction (Höppner), we mirror human biomechanics. Therefore, we can maintain constant force while varying compliance to match task demands, or also for hold objects firmly with low force, such as with an egg, and squeeze them with low stiffness, such as with a stress ball.

Notes: *define between case study and what it is. We can think everything as tuning the slope. When we work in "position control", the slope between force and reference is very high: extreme rigidity. When we work in "force control", the slope is usually low: extreme flexibility. On the figure, we show again the force/displacement graph with force and position control through stiffness and then we connect to the two cases, so showing that we can "move" in an agile way there.*

Discussion

Discussion text.

Why do we have body compliance (e.g., silicon, VSA, tendons, springs, etc) and control compliance? Control compliance can be controlled but is not fast as the body one, since we need some time to close the loop. Moreover, it can be distributed easily and does not suffer from sensor or actuator errors.

Methods

Experimental setup

1. Full hand platform: description of fabrication/setup
2. Picture of single finger platform and relationship

Tendon-driven compliant hand

Characteristics of the finger and hand, high-level description, and how the finger is modeled kinematically (MCP, PIP, DIP), and modeling of the hand.

1. Finger kinematics
2. Friction compensation

VMC modeling for finger and hand

1. VMC for finger: show finger space, task space, carts, the union of both, and so on
2. VMC for the hand, how to model it, kinematics, transformations, and so on

Passivity and stiffness control through VMC

All theoretical formulation, *frictionless point contact*.

Force sensing and control through VMC

How to relate efficiency and how to use, then:

1. Backpropagation of stiffness for force and position control
2. Backpropagation of target for force and stiffness control

Temporary methods:

Finger-space formulation

We will show here part of the paper *Methods*, namely how the stiffness of VMC can be translated in system stiffness and how this analysis relates to the force applied by the finger (and therefore by the hand) to external objects. Let us denote the fingertip position as:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Assume the VMC framework is the finger space one, namely in *finger-space*. This means putting (as in the first experiments) virtual elements in the MCP, PIP, DIP joints. Let us denote these coordinates as:

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_{MCP} \\ \theta_{PIP} \\ \theta_{DIP} \end{bmatrix}$$

The corresponding stiffness matrix is:

$$\mathbf{K}_{\theta} = \begin{bmatrix} k_{MCP} & 0 & 0 \\ 0 & k_{PIP} & 0 \\ 0 & 0 & k_{DIP} \end{bmatrix}$$

The tendon transmission efficiency matrix, diagonal in actuator space, accounts for friction losses in the tendon routing and for the power dissipated in the spring due to compliance (in η_2):

$$\boldsymbol{\eta} = \begin{bmatrix} \eta_1 & 0 \\ 0 & \eta_2 \end{bmatrix}, \quad \eta_i \in (0, 1]$$

The applied torque is therefore:

$$\boldsymbol{\tau}_{\theta} = \mathbf{K}_{\theta} \cdot (\boldsymbol{\theta}_{ref} - \boldsymbol{\theta})$$

We want to compute the force at the fingertip according to the Principle of Virtual Work (PVW):

$$\dot{\boldsymbol{\theta}}^T \cdot \boldsymbol{\tau}_\theta = \dot{\mathbf{x}}^T \cdot \mathbf{f}_x$$

In this equation, $\mathbf{f}_x$ is the generalized force on the fingertip that would produce the same effect of $\boldsymbol{\tau}_\theta$ on the robot. If we had the possibility of measuring this force with a "spiderweb" of infinite load cells in all directions, we would see exactly this force. However, when contact happens, we can model it with a *frictionless point contact*, ignoring all the effects related to the friction cone. Given the contact plane versor identified by $\mathbf{n}$, and the corresponding orthogonal projection matrix $\mathbf{P} = \mathbf{n}\mathbf{n}^T$:

$$\dot{\boldsymbol{\theta}}^T \cdot \boldsymbol{\tau}_\theta = (\mathbf{P}\dot{\mathbf{x}})^T \cdot \mathbf{f}_x$$

In this case, forces only happen along $\mathbf{n}$. Being:

$$\dot{\mathbf{x}} = \mathbf{J}_x \cdot \dot{\mathbf{q}}$$

$$\dot{\boldsymbol{\theta}} = \mathbf{J}_\theta \cdot \dot{\mathbf{q}}$$

Substituting the Jacobian relations into the projected PVW equation:

$$(\mathbf{J}_\theta \cdot \dot{\mathbf{q}})^T \cdot \boldsymbol{\tau}_\theta = (\mathbf{P} \cdot \mathbf{J}_x \cdot \dot{\mathbf{q}})^T \cdot \mathbf{f}_x$$

$$\mathbf{J}_\theta^T \cdot \boldsymbol{\tau}_\theta = (\mathbf{P} \cdot \mathbf{J}_x)^T \cdot \mathbf{f}_x$$

However, the virtual torques $\boldsymbol{\tau}_\theta$ are not fully transmitted to the joints due to tendon routing friction. Accounting for the transmission efficiency $\boldsymbol{\eta}$ in actuator space:

$$\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \cdot \boldsymbol{\tau}_\theta = (\mathbf{P} \cdot \mathbf{J}_x)^T \cdot \mathbf{f}_x$$

Solving using the pseudoinverse:

$$\boxed{\mathbf{f}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \cdot \mathbf{K}_\theta \cdot (\boldsymbol{\theta}_{ref} - \boldsymbol{\theta})}$$

This is the contact force at the fingertip, energetically consistent with the virtual torques through PVW. When no contact is present, $\mathbf{P} = \mathbf{I}$ and $\mathbf{f}_x$ represents the full generalized force.

The corresponding stiffness is obtained by differentiating the virtual work equality $\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \boldsymbol{\tau}_\theta = (\mathbf{P} \cdot \mathbf{J}_x)^T \mathbf{f}_x$ with respect to $\mathbf{q}$. Denoting $d\boldsymbol{\tau}_\theta = -\mathbf{K}_\theta d\boldsymbol{\theta}$ and $d\mathbf{f}_x = -\mathbf{K}_x d(\mathbf{P}\mathbf{x})$:

$$-(\mathbf{P} \cdot \mathbf{J}_x)^T \mathbf{K}_x (\mathbf{P} \cdot \mathbf{J}_x) + \frac{\partial(\mathbf{P} \cdot \mathbf{J}_x)^T}{\partial \mathbf{q}} \mathbf{f}_x = -\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \mathbf{K}_\theta \mathbf{J}_\theta + \frac{\partial(\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T)}{\partial \mathbf{q}} \boldsymbol{\tau}_\theta$$

Solving for $\mathbf{K}_x$, with the virtual torque $\boldsymbol{\tau}_\theta = \mathbf{K}_\theta \cdot (\boldsymbol{\theta}_{ref} - \boldsymbol{\theta})$ known from the virtual model:

$$\mathbf{K}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \left[\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \mathbf{K}_\theta \mathbf{J}_\theta + \frac{\partial(\mathbf{P} \cdot \mathbf{J}_x)^T}{\partial \mathbf{q}} \cdot \mathbf{f}_x - \frac{\partial(\boldsymbol{\eta} \cdot \mathbf{J}_\theta^T)}{\partial \mathbf{q}} \cdot \boldsymbol{\tau}_\theta \right] ((\mathbf{P} \cdot \mathbf{J}_x)^*)$$

This expression is exact: the geometric correction terms involve the Hessians of the output and virtual-element Jacobians, weighted respectively by the external force $\mathbf{f}_x$ and the virtual torque $\boldsymbol{\tau}_\theta$, consistent with the conservative congruence transformation (CCT). Note that an alternative formulation, obtained by substituting $\boldsymbol{\tau}_\theta = (\mathbf{J}_\theta^*)^T \cdot \boldsymbol{\eta}^{-1} \cdot (\mathbf{P} \cdot \mathbf{J}_x)^T \mathbf{f}_x$ to express $\mathbf{K}_x$ as a function of $\mathbf{f}_x$ alone, is only valid when $\mathbf{J}_d$ has full column rank (i.e., $\dim(\mathbf{d}) \leq \dim(\mathbf{q})$). In the general case, the pseudoinverse recovers only the minimum-norm virtual force among the infinitely many $\boldsymbol{\tau}_d$ that produce the same tip force $\mathbf{f}_x$, discarding the null-space component. Since the actual virtual force is uniquely determined by the virtual model as $\boldsymbol{\tau}_d = \mathbf{K}_d \cdot (\mathbf{d}_{ref} - \mathbf{d})$, the formulation above, which uses $\boldsymbol{\tau}_\theta$ directly, is both exact and general. Linearized to first order (neglecting Jacobian derivative terms), as an extension of Salisbury's famous paper:

$$\boxed{\mathbf{K}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_\theta^T \cdot \mathbf{K}_\theta \cdot \mathbf{J}_\theta \cdot (\mathbf{P} \cdot \mathbf{J}_x)^*}$$

General formulation The derivation generalizes to *any* set of virtual elements. Let $\mathbf{d}$ denote a vector of generalized Lagrange variables in which virtual elements are defined (e.g., $\mathbf{d} = [\theta_{MCP}, \theta_{PIP}, \theta_{DIP}, x_1, x_2]^T$ for combined joint-space and task-space springs. Let $\mathbf{K}_d$ be the associated stiffness matrix and $\mathbf{J}_d = \partial \mathbf{d} / \partial \mathbf{q}$ the corresponding Jacobian. The projection matrix $\mathbf{P}$ selects which components of the output velocity $\dot{\mathbf{x}}$ perform virtual work: $\mathbf{P} = \mathbf{I}$ recovers the full generalized force in all directions, while $\mathbf{P} = \mathbf{n}\mathbf{n}^T$ restricts to the normal contact force along $\mathbf{n}$. By the same PVW argument, and including the tendon transmission efficiency η :

$$\mathbf{f}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \eta \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d \cdot (\mathbf{d}_{\text{ref}} - \mathbf{d})$$

If $\mathbf{K}_d$ is decoupled (diagonal), this decomposes as a sum of independent virtual contributions:

$$\mathbf{f}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \eta \cdot \sum_i \mathbf{J}_{d_i}^T k_i (d_{i,\text{ref}} - d_i)$$

By the same differentiation of virtual work, the exact stiffness is:

$$\mathbf{K}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \left[\eta \cdot \mathbf{J}_d^T \mathbf{K}_d \mathbf{J}_d + \frac{\partial (\mathbf{P} \cdot \mathbf{J}_x)^T}{\partial \mathbf{q}} \cdot \mathbf{f}_x - \frac{\partial (\eta \cdot \mathbf{J}_d^T)}{\partial \mathbf{q}} \cdot \boldsymbol{\tau}_d \right] ((\mathbf{P} \cdot \mathbf{J}_x)^*)$$

where $\boldsymbol{\tau}_d = \mathbf{K}_d \cdot (\mathbf{d}_{\text{ref}} - \mathbf{d})$ is the generalized virtual force, directly available from the virtual model. This formulation is exact regardless of the dimension of $\mathbf{d}$ relative to $\mathbf{q}$, as it avoids the pseudoinverse reconstruction of the virtual forces.

It is important to notice that, for our derivatives, we need to assume constant efficiency, which quantifies in this case the *average* effect of losses on the given system. Therefore, we can estimate it in a data-driven way. Keep in mind that the motor friction must be compensated in any case a priori, since we are quantifying the effect of *transmission* of the system (and not of the electric motor itself, which is already considered by the constant k_T).

Theorem 1 (VMC Stiffness Composition). *Consider a tendon-driven robotic system with:*

1. configuration $\mathbf{q} \in \mathbb{R}^{n_q}$,

2. output Jacobian $\mathbf{J}_x = \partial \mathbf{x} / \partial \mathbf{q}$,
3. output projection $\mathbf{P}$ selecting the directions along which virtual work is performed,
4. tendon transmission efficiency $\boldsymbol{\eta} \in \mathbb{R}^{n_q \times n_q}$, diagonal with $\eta_i \in (0, 1]$, constant w.r.t. $\mathbf{q}$,
5. a set of virtual elements defined in generalized coordinates $\mathbf{d}$ with Jacobian $\mathbf{J}_d = \partial \mathbf{d} / \partial \mathbf{q}$ and positive semi-definite stiffness matrix $\mathbf{K}_d$.

The output-space stiffness, to first order, is:

$$\mathbf{K}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d \cdot \mathbf{J}_d \cdot (\mathbf{P} \cdot \mathbf{J}_x)^*$$

If the virtual elements are decoupled:

$$\mathbf{K}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \left(\sum_i \mathbf{J}_{d_i}^T k_i \mathbf{J}_{d_i} \right) \cdot (\mathbf{P} \cdot \mathbf{J}_x)^*$$

The force and stiffness at any output point $\mathbf{x}$ are fully determined by the virtual element parameters, the kinematic structure, the tendon transmission efficiency, and the output projection, without requiring dynamic models. Virtual contributions from any coordinate space compose additively through PVW.

How to compute the compliance of the system and how to use that in the stiffness feedback system for force tuning

Once we proved that we can map stiffness and force output to VMC control input, we must find a way to vary stiffness at the Lagrangian coordinates $\mathbf{d}$ to achieve a desired force. In other words, to find a model - either feedforward or feedback - that allows to adapt force accordingly to Theorem 1. Let us consider the following equation:

$$\mathbf{f}_x = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d \cdot (\mathbf{d}_{\text{ref}} - \mathbf{d})$$

Intuitively, solving for $\mathbf{K}_d$ should solve the issue. But this approach is wrong. In fact, as soon as $\mathbf{K}_d$ changes, the state of the manipulator changes as well, i.e., $\mathbf{d}$ varies. The problem must be faced with another perspective.

In any case, we want to tune the stiffness $\mathbf{K}_d$ of the VMC online to minimize the difference between the reference force and the actual one. This means minimizing the scalar loss:

$$\mathcal{L} = \frac{1}{2}(\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})^T(\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})$$

To derive the update rule using standard matrix calculus, let us define the task-space mapping matrix $\mathbf{A} = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T$ and the virtual deflection $\mathbf{u} = \mathbf{d}_{\text{ref}} - \mathbf{d}$, such that the predicted force is compactly written as $\mathbf{f}_x = \mathbf{A}\mathbf{K}_d\mathbf{u}$.

The gradient of the scalar loss $\mathcal{L}$ with respect to the stiffness matrix $\mathbf{K}_d$ is given by the outer product:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{K}_d} = \mathbf{A}^T(\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})\mathbf{u}^T$$

Substituting the original terms back, we can update the stiffness matrix online via a direct rank-1 gradient descent step:

$$\mathbf{K}_d^{(i+1)} = \mathbf{K}_d^{(i)} - \alpha [\mathbf{J}_d \cdot \boldsymbol{\eta}^T \cdot ((\mathbf{P} \cdot \mathbf{J}_x)^*)] (\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})(\mathbf{d}_{\text{ref}} - \mathbf{d})^T$$

where $\alpha > 0$ is the learning rate, and i is the iteration index of the gradient descent.

The beauty in all of that also lies in the fact that the robot can be seen, with an effort of imagination, as a *reservoir computing* case. In fact, given a desired output of the system (the tip force), we are able to tune a linear layer of the system (the control stiffness) with gradient descent.

Force tracking via reference modulation

An alternative to modulating the stiffness matrix $\mathbf{K}_d$ online is to keep it fixed and instead adapt the virtual reference $\mathbf{d}_{\text{ref}}$. Starting from the same force expression:

$$\mathbf{f}_x = \underbrace{((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d}_{\mathbf{S}} \cdot (\mathbf{d}_{\text{ref}} - \mathbf{d})$$

the force-reference sensitivity is simply:

$$\frac{\partial \mathbf{f}_x}{\partial \mathbf{d}_{\text{ref}}} = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d = \mathbf{S}$$

The gradient of the loss $\mathcal{L} = \frac{1}{2} \|\mathbf{f}_x - \mathbf{f}_{x,\text{ref}}\|^2$ with respect to $\mathbf{d}_{\text{ref}}$ is therefore:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{d}_{\text{ref}}} = \mathbf{S}^T (\mathbf{f}_x - \mathbf{f}_{x,\text{ref}}) = \mathbf{K}_d^T \cdot \mathbf{J}_d \cdot \boldsymbol{\eta}^T \cdot ((\mathbf{P} \cdot \mathbf{J}_x)^*) \cdot (\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})$$

yielding the update rule:

$$\mathbf{d}_{\text{ref}}^{(i+1)} = \mathbf{d}_{\text{ref}}^{(i)} - \alpha \mathbf{K}_d^T \cdot \mathbf{J}_d \cdot \boldsymbol{\eta}^T \cdot ((\mathbf{P} \cdot \mathbf{J}_x)^*) \cdot (\mathbf{f}_x - \mathbf{f}_{x,\text{ref}})$$

where $\alpha > 0$ is the learning rate. Compared to the stiffness-modulation scheme, this formulation operates directly on a vector in $\mathbb{R}^{n_d}$ rather than on the vectorized matrix in $\mathbb{R}^{n_d^2}$, avoiding Kronecker products entirely. Moreover, since $\mathbf{f}_x$ depends linearly on $\mathbf{d}_{\text{ref}}$ at frozen configuration, a single step with $\alpha = 1$ would be exact if $\mathbf{d}$ remained constant. In practice, changing $\mathbf{d}_{\text{ref}}$ also shifts $\mathbf{d}$ through the compliant dynamics, so iterative updates are still required, but convergence is expected to be faster than in the $\mathbf{K}_d$ -modulation case due to the milder coupling. In the reservoir computing analogy, this corresponds to tuning the *bias* of the linear readout layer rather than its *weights*. More importantly, as noted in some biomechanical papers, we are able to *decouple* stiffness and force with this strategy, as it happens in human-like control. This means that we can independently tune stiffness and force, so due to the variation of the spring itself we have full control on two variables between stiffness, force, and pose.

Feed-forward stiffness inversion

The first-order task-space stiffness provided by a virtual element is given by the congruence transformation $\mathbf{K}_x = \mathbf{L} \mathbf{K}_d \mathbf{R}$. Vectorizing this matrix equation via the standard identity $\text{vec}(\mathbf{AXB}) = (\mathbf{B}^T \otimes \mathbf{A}) \text{vec}(\mathbf{X})$ yields the linear mapping:

$$\text{vec}(\mathbf{K}_x) = (\mathbf{R}^T \otimes \mathbf{L}) \text{vec}(\mathbf{K}_d)$$

When a single virtual-element group is considered, the minimum-norm virtual stiffness $\mathbf{K}_d^{ff}$ that best approximates a desired task-space profile $\mathbf{K}_{x,\text{des}}$ in a least-squares sense is given by:

$$\text{vec}(\mathbf{K}_d^{ff}) = (\mathbf{R}^T \otimes \mathbf{L})^+ \text{vec}(\mathbf{K}_{x,\text{des}})$$

where $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse.

However, when multiple virtual-element groups contribute additively to the stiffness of a single end-effector (e.g., combining different joint-space spring groups for a single finger), the total task-space stiffness is the sum of their individual contributions: $\mathbf{K}_x = \sum_g \mathbf{L}_g \mathbf{K}_{d_g} \mathbf{R}_g$. Because the unknown virtual stiffness matrices are bounded by distinct kinematic mappings, a joint minimum-norm inversion must be solved simultaneously across all groups. By defining $\mathbf{M}_g = \mathbf{R}_g^T \otimes \mathbf{L}_g$, the linear maps are stacked horizontally into a single generalized inversion:

$$\begin{bmatrix} \text{vec}(\mathbf{K}_{d_1}) \\ \vdots \\ \text{vec}(\mathbf{K}_{d_G}) \end{bmatrix}^{ff} = [\mathbf{M}_1 \quad \cdots \quad \mathbf{M}_G]^+ \text{vec}(\mathbf{K}_{x,\text{des}})$$

This joint inversion evaluates exactly when the target lies in the achievable range; otherwise, it returns the minimum-norm least-squares approximation. It can be evaluated at each time step to optimally distribute the required stiffness across all available virtual spring groups to track a time-varying profile.

Positive definiteness of the stiffness matrix

Theorem 2 (One-step positive definiteness). *Let*

$$\mathbf{K}_d^{(i+1)} = \mathbf{K}_d^{(i)} - \alpha e_i \mathbf{v}_i \mathbf{u}_i^T$$

with

$$\mathbf{A}_i = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T, \quad \mathbf{u}_i = \mathbf{d}_{\text{ref}} - \mathbf{d}_i, \quad \mathbf{v}_i = \mathbf{A}_i^T \mathbf{n}, \quad e_i = f_i - f_{\text{ref}}.$$

Assume $\mathbf{K}_d^{(i)}$ is positive definite and define

$$\lambda_i = \lambda_{\min} \left(\frac{1}{2} \left(\mathbf{K}_d^{(i)} + \mathbf{K}_d^{(i)T} \right) \right) > 0, \quad \beta_i = |e_i| \|\mathbf{u}_i\| \|\mathbf{v}_i\|.$$

If

$$0 < \alpha < \frac{\lambda_i}{\beta_i},$$

then $\mathbf{K}_d^{(i+1)}$ is positive definite.

Proof. For any **unit** vector $\mathbf{x}$,

$$\mathbf{x}^T \mathbf{K}_d^{(i+1)} \mathbf{x} = \mathbf{x}^T \mathbf{K}_d^{(i)} \mathbf{x} - \alpha e_i (\mathbf{x}^T \mathbf{v}_i) (\mathbf{u}_i^T \mathbf{x}).$$

Since $\mathbf{x}^T \mathbf{K}_d^{(i)} \mathbf{x} = \mathbf{x}^T \frac{1}{2} (\mathbf{K}_d^{(i)} + \mathbf{K}_d^{(i)T}) \mathbf{x} \geq \lambda_i$, and by Cauchy–Schwarz,

$$|(\mathbf{x}^T \mathbf{v}_i) (\mathbf{u}_i^T \mathbf{x})| \leq \|\mathbf{v}_i\| \|\mathbf{u}_i\|,$$

we obtain

$$\mathbf{x}^T \mathbf{K}_d^{(i+1)} \mathbf{x} \geq \lambda_i - \alpha \beta_i.$$

If $\alpha < \lambda_i / \beta_i$, then $\mathbf{x}^T \mathbf{K}_d^{(i+1)} \mathbf{x} > 0$ for all $\|\mathbf{x}\| = 1$, hence $\mathbf{K}_d^{(i+1)}$ is positive definite. $\square$

Lemma 1 (Finite-horizon bound). *Assume $\lambda_i \geq \underline{\lambda} > 0$ and $\beta_i \leq \bar{\beta}$ for all $i = 0, \dots, N-1$.*

Then any

$$0 < \alpha < \frac{\underline{\lambda}}{\bar{\beta}}$$

guarantees $\mathbf{K}_d^{(i)}$ positive definite for all $i = 0, \dots, N$.

Proof. By Theorem 2, the one-step condition

$$0 < \alpha < \frac{\lambda_i}{\beta_i}$$

guarantees $\mathbf{K}_d^{(i+1)}$ is positive definite whenever $\mathbf{K}_d^{(i)}$ is. Under the assumptions $\lambda_i \geq \underline{\lambda}$ and $\beta_i \leq \overline{\beta}$, we have

$$\frac{\lambda_i}{\beta_i} \geq \frac{\underline{\lambda}}{\overline{\beta}}.$$

Thus any constant step size

$$0 < \alpha < \frac{\underline{\lambda}}{\overline{\beta}}$$

satisfies the one-step condition for every $i = 0, \dots, N - 1$. Applying the one-step result iteratively yields $\mathbf{K}_d^{(i)}$ positive definite for all $i = 0, \dots, N$. $\square$

Theorem 3 (Stability of stiffness backpropagation). *Assume a frictionless point contact with $\mathbf{P} = \mathbf{n}\mathbf{n}^T$, and define*

$$\mathbf{A}_i = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T, \quad \mathbf{u}_i = \mathbf{d}_{\text{ref}} - \mathbf{d}_i, \quad \mathbf{v}_i = \mathbf{A}_i^T \mathbf{n}.$$

The stiffness step

$$\mathbf{K}_d^{(i+1)} = \mathbf{K}_d^{(i)} - \alpha e_i \mathbf{v}_i \mathbf{u}_i^T, \quad e_i = f_i - f_{\text{ref}}, \quad f_i = \mathbf{v}_i^T \mathbf{K}_d^{(i)} \mathbf{u}_i$$

produces the error recursion

$$e_{i+1} = (1 - \alpha \gamma_i) e_i, \quad \gamma_i = \|\mathbf{v}_i\|^2 \|\mathbf{u}_i\|^2.$$

Therefore the force error is stable and convergent if and only if

$$0 < \alpha < \frac{2}{\gamma_i},$$

and monotone if $0 < \alpha < 1/\gamma_i$ (if $\gamma_i = 0$, the error is unchanged).

Proof. Using the update,

$$f_{i+1} = \mathbf{v}_i^T \mathbf{K}_d^{(i+1)} \mathbf{u}_i = f_i - \alpha e_i \|\mathbf{v}_i\|^2 \|\mathbf{u}_i\|^2,$$

so $e_{i+1} = e_i - \alpha \gamma_i e_i = (1 - \alpha \gamma_i) e_i$. Stability requires $|1 - \alpha \gamma_i| < 1$, giving $0 < \alpha < 2/\gamma_i$. $\square$

Theorem 4 (Stability of reference backpropagation). *Let*

$$\mathbf{S}_i = ((\mathbf{P} \cdot \mathbf{J}_x)^*)^T \cdot \boldsymbol{\eta} \cdot \mathbf{J}_d^T \cdot \mathbf{K}_d, \quad \mathbf{f}_{x,i} = \mathbf{S}_i(\mathbf{d}_{\text{ref}}^{(i)} - \mathbf{d}_i),$$

and define the reference step

$$\mathbf{d}_{\text{ref}}^{(i+1)} = \mathbf{d}_{\text{ref}}^{(i)} - \alpha \mathbf{S}_i^T (\mathbf{f}_{x,i} - \mathbf{f}_{x,\text{ref}}).$$

Then the error recursion is

$$\mathbf{e}_{i+1} = (\mathbf{I} - \alpha \mathbf{S}_i \mathbf{S}_i^T) \mathbf{e}_i, \quad \mathbf{e}_i = \mathbf{f}_{x,i} - \mathbf{f}_{x,\text{ref}}.$$

Thus the update is stable and convergent if and only if

$$0 < \alpha < \frac{2}{\sigma_{\max}^2(\mathbf{S}_i)}.$$

Proof. Substituting the update into $\mathbf{f}_{x,i}$ gives $\mathbf{e}_{i+1} = (\mathbf{I} - \alpha \mathbf{S}_i \mathbf{S}_i^T) \mathbf{e}_i$. Stability requires the spectral radius of $(\mathbf{I} - \alpha \mathbf{S}_i \mathbf{S}_i^T)$ to be < 1 , which is equivalent to $0 < \alpha < 2/\sigma_{\max}^2(\mathbf{S}_i)$. $\square$

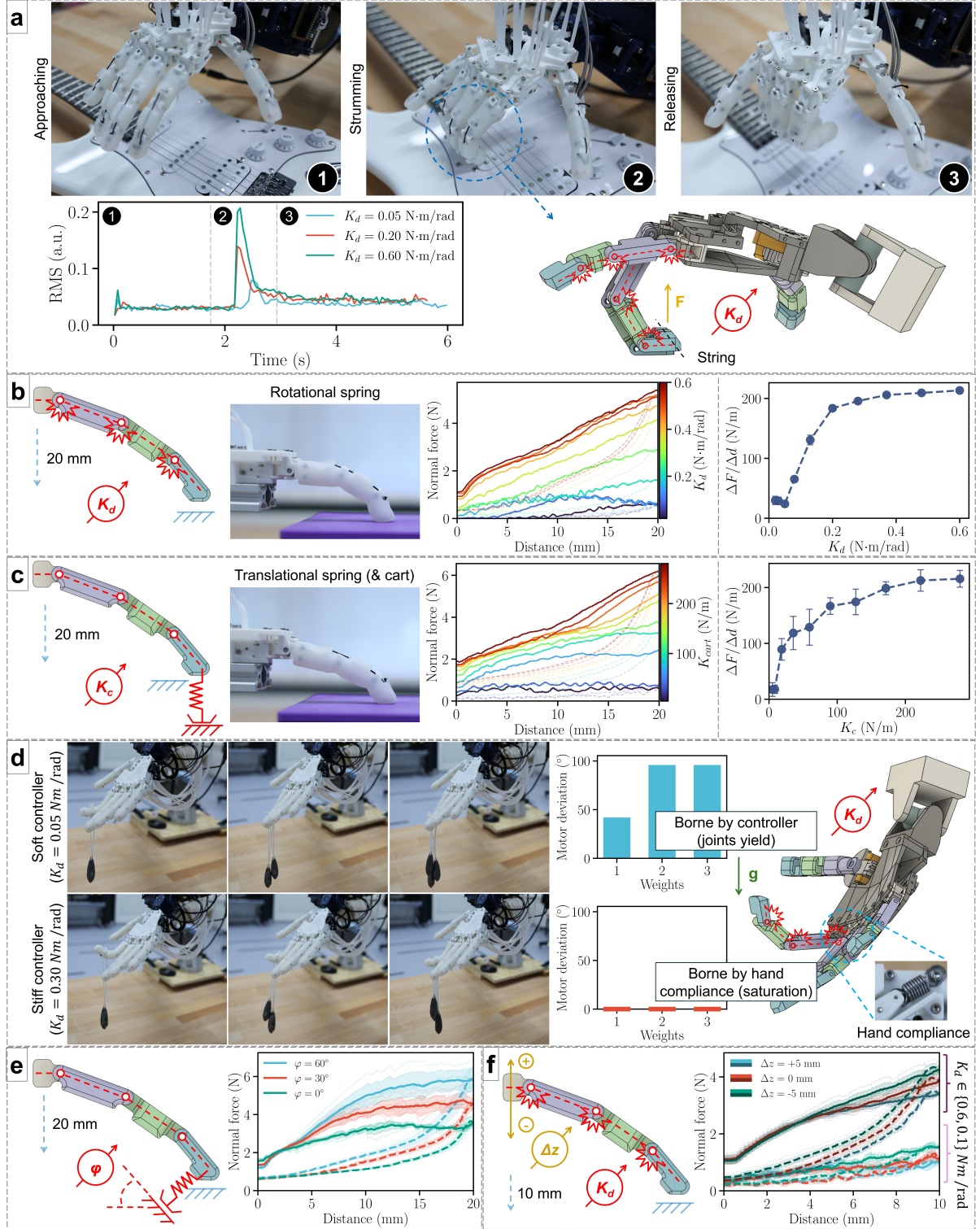


Figure 3: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

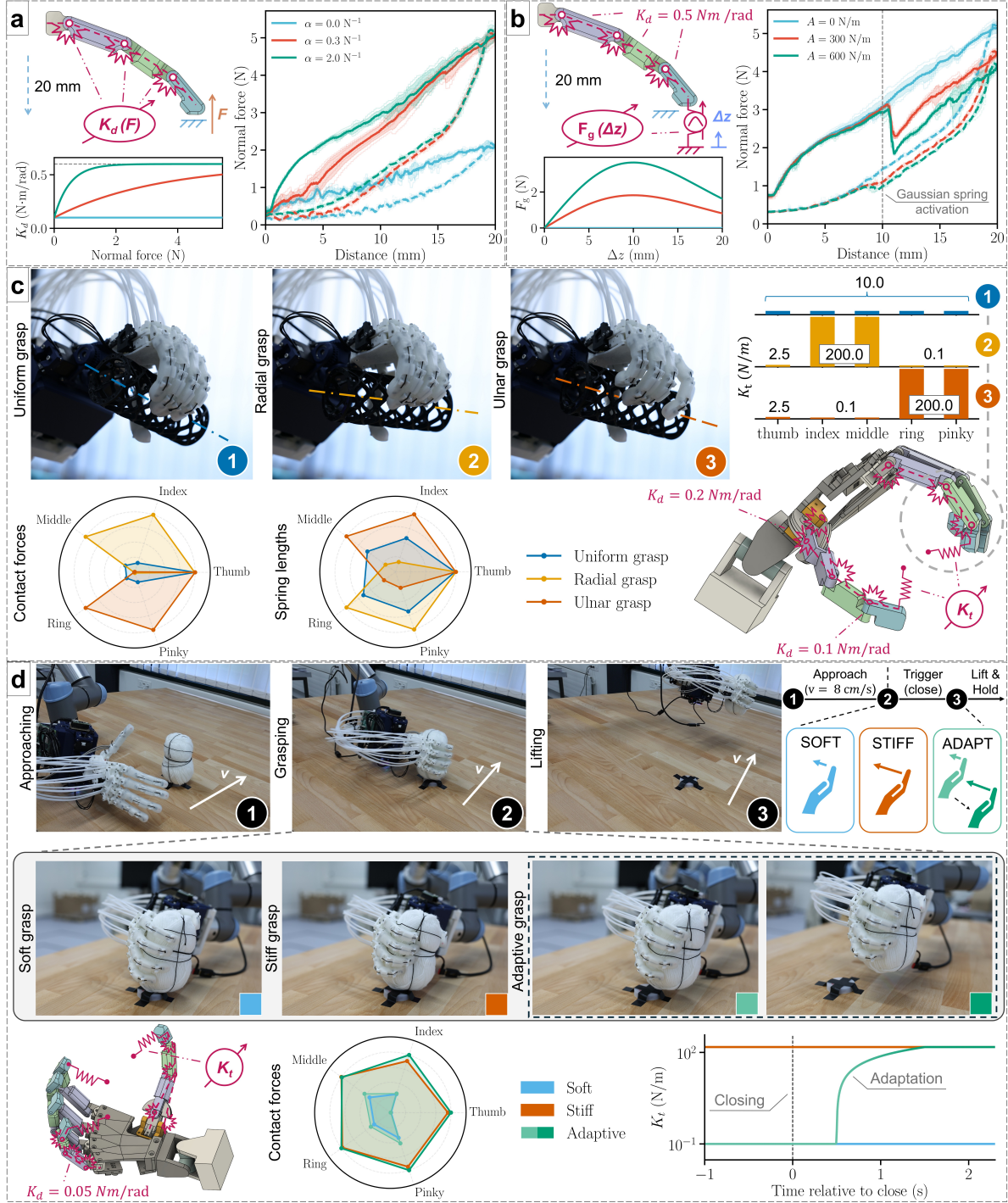


Figure 4: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

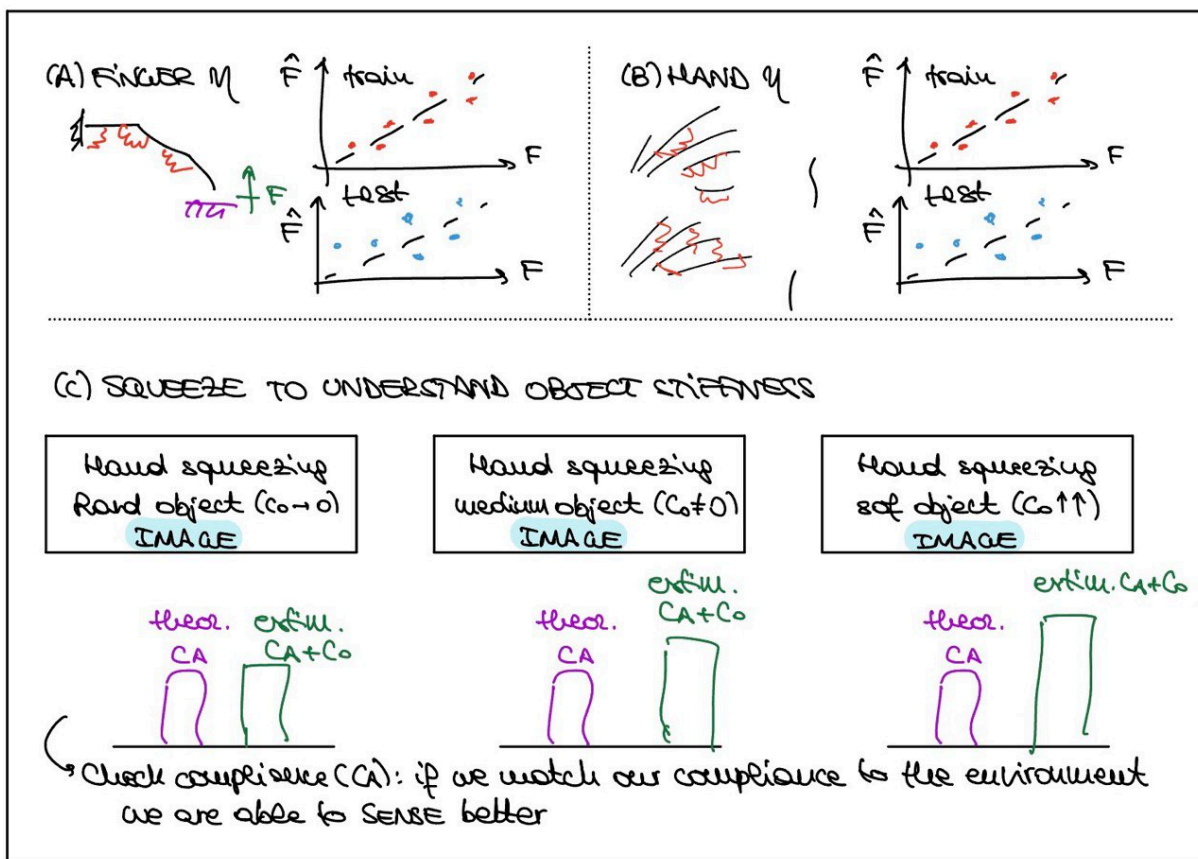


Figure 5: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

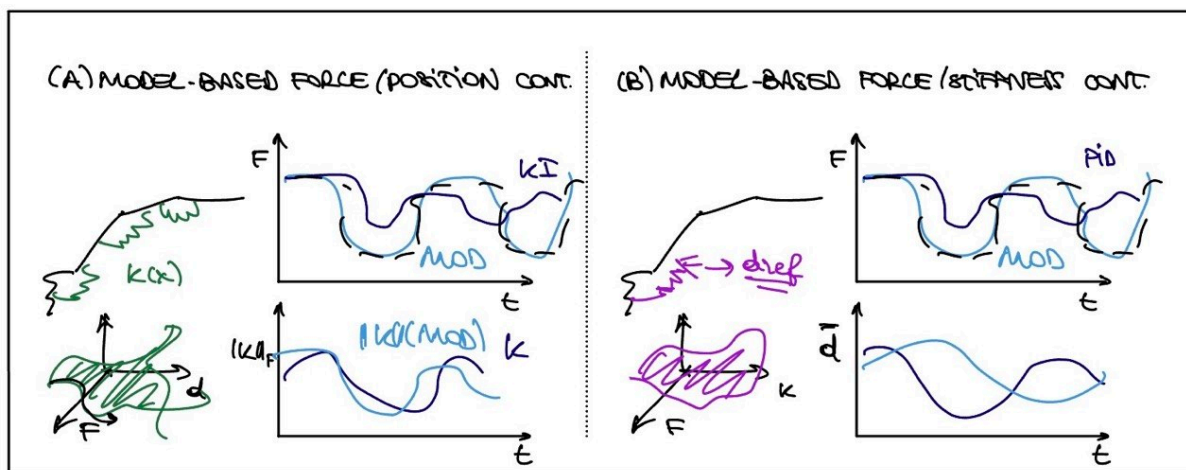


Figure 6: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

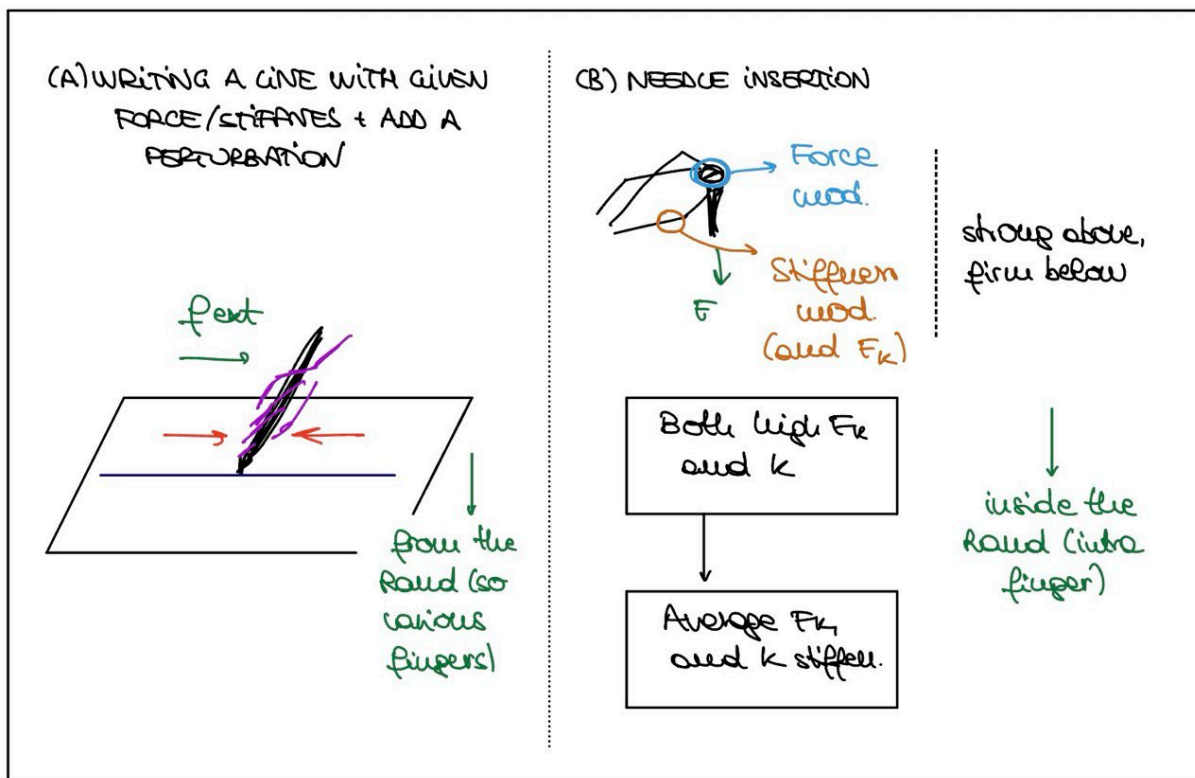


Figure 7: **Big title.** (a) First.(b) Second one. (c-d) Third and fourth.

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Competing interests

The authors declare that they have no competing interests.

Author contributions

Conceptualization: L.V., F.F., and J.H. **Methodology:** all authors. **Investigation:** S.P. and O. V. **Data analysis:** S.P. **Visualization:** S.P. **Supervision:** J.T.B.O. **Writing:** S.P. and J.T.B.O. **Review and editing:** all authors.

Fundings

Future work:

1. Add dampers since we are thinking in terms of static, but we must also think in terms of dynamics.
2. Co-design for control.
3. Replaceable fingers.
4. Estimation of task-specific VMC parameters for learning.

Data and materials availability

Put data and code.

References

- [1] Burdet, E., Osu, R., Franklin, D. W., Milner, T. E. & Kawato, M. The Central Nervous System Stabilizes Unstable Dynamics by Learning Optimal Impedance. *Nature* **414**, 446–449 (2001).
- [2] Tee, K. P., Franklin, D. W., Kawato, M., Milner, T. E. & Burdet, E. Concurrent Adaptation of Force and Impedance in the Redundant Muscle System. *Biological Cybernetics* **102**, 31–44 (2010).
- [3] Höppner, H., Große-Dunker, M., Stillfried, G., Bayer, J. & Van Der Smagt, P. Key Insights into Hand Biomechanics: Human Grip Stiffness Can Be Decoupled from Force by Cocontraction and Predicted from Electromyography. *Frontiers in Neurorobotics* **11**, 17 (2017).
- [4] Milner, T. E. & Cloutier, C. Compensation for Mechanically Unstable Loading in Voluntary Wrist Movement. *Experimental Brain Research* **94** (1993).
- [5] Billard, A. & Kragic, D. Trends and Challenges in Robot Manipulation. *Science* **364**, eaat8414 (2019).
- [6] Gilday, K., Sirithunge, C., Iida, F. & Hughes, J. Embodied Manipulation with Past and Future Morphologies through an Open Parametric Hand Design. *Science Robotics* (2025).
- [7] Gilday, K., Pyeon, D., Dhanush, S., Cho, K. & Hughes, J. Exploiting Passive Behaviours for Diverse Musical Playing Using the Parametric Hand. *Frontiers in Robotics and AI* **11**, 1463744 (2024).
- [8] Hughes, J. A. E., Maiolino, P. & Iida, F. An Anthropomorphic Soft Skeleton Hand Exploiting Conditional Models for Piano Playing. *Science Robotics* **3**, eaau3098 (2018).

- [9] Stella, F., Achkar, M. M., Della Santina, C. & Hughes, J. Synergy-Based Robotic Quadruped Leveraging Passivity for Natural Intelligence and Behavioural Diversity. *Nature Machine Intelligence* **7**, 386–399 (2025).
- [10] Piazza, C., Grioli, G., Catalano, M. G. & Bicchi, A. A Century of Robotic Hands. *Annual Review of Control, Robotics, and Autonomous Systems* **2**, 1–32 (2019).
- [11] Bicchi, A. & Kumar, V. Robotic Grasping and Contact: A Review. *Proceedings 2000 ICRA. Millennium Conference. IEEE International Conference on Robotics and Automation. Symposia Proceedings* 348–353 (IEEE, 2000).
- [12] Willems, J. C. Dissipative Dynamical Systems. *European Journal of Control* **13**, 134–151 (2007).
- [13] Dickinson, M. H., Farley, C. T., Full, R. J., Koehl, M. A. R., Kram, R. & Lehman, S. How Animals Move: An Integrative View. *Science* **288**, 100–106 (2000).
- [14] Ijspeert, A. J. & Daley, M. A. Integration of Feedforward and Feedback Control in the Neuromechanics of Vertebrate Locomotion: A Review of Experimental, Simulation and Robotic Studies. *Journal of Experimental Biology* **226**, jeb245784 (2023).
- [15] Siciliano, B. & Khatib, O. *Springer Handbook of Robotics* (Springer International Publishing, 2016).
- [16] Scimeca, L., Ng, C. & Iida, F. Gaussian Process Inference Modelling of Dynamic Robot Control for Expressive Piano Playing. *PLOS ONE* **15**, e0237826 (2020).
- [17] Zeng, C., Li, S., Chen, Z., Yang, C., Sun, F. & Zhang, J. Multifingered Robot Hand Compliant Manipulation Based on Vision-Based Demonstration and Adaptive Force Control. *IEEE Transactions on Neural Networks and Learning Systems* **34**, 5452–5463 (2023).

- [18] SaLoutos, A., Kim, H., Stanger-Jones, E., Guo, M. & Kim, S. Towards Robust Autonomous Grasping with Reflexes Using High-Bandwidth Sensing and Actuation. *2023 IEEE International Conference on Robotics and Automation (ICRA)* 10254–10260 (IEEE, 2023).
- [19] Ye, Q., Liu, Q., Wang, S., Chen, J., Cui, Y., Jin, K. et al. Visual-Tactile Pretraining and On-line Multitask Learning for Humanlike Manipulation Dexterity. *Science Robotics* (2026).
- [20] Yang, Y., Li, Y. & Chen, Y. Principles and Methods for Stiffness Modulation in Soft Robot Design and Development. *Bio-Design and Manufacturing* **1**, 14–25 (2018).
- [21] Wolf, S., Grioli, G., Eiberger, O., Friedl, W., Grebenstein, M., Hoppner, H. et al. Variable Stiffness Actuators: Review on Design and Components. *IEEE/ASME Transactions on Mechatronics* **21**, 2418–2430 (2016).
- [22] Obayashi, N., Junge, K., Singh, P. & Hughes, J. Online Hydraulic Stiffness Modulation of a Soft Robotic Fish Tail for Improved Thrust and Efficiency. *Soft Robotics* soro.2024.0030 (2024).
- [23] Yan, J., Xu, Z., Shi, P. & Zhao, J. A Human-Inspired Soft Finger with Dual-Mode Morphing Enabled by Variable Stiffness Mechanism. *Soft Robotics* **9**, 399–411 (2022).
- [24] Yang, Y., Chen, Y., Li, Y., Wang, Z. & Li, Y. Novel Variable-Stiffness Robotic Fingers with Built-In Position Feedback. *Soft Robotics* **4**, 338–352 (2017).
- [25] Junge, K., Noda, T. & Hughes, J. Mod-VSA: A Modular Variable Stiffness Actuator for Multi-fingered Robotic Hands. *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)* 1–6 (IEEE, 2025).

- [26] Song, P., Yu, Y. & Zhang, X. Impedance Control of Robots: An Overview. *2017 2nd International Conference on Cybernetics, Robotics and Control (CRC)* 51–55 (IEEE, 2017).
- [27] Albu-Schaffer, A. & Hirzinger, G. Cartesian Impedance Control Techniques for Torque Controlled Light-Weight Robots. *Proceedings 2002 IEEE International Conference on Robotics and Automation* 657–663 (IEEE, 2002).
- [28] Ott, C., Albu-Schaffer, A., Kugi, A. & Hirzinger, G. On the Passivity-Based Impedance Control of Flexible Joint Robots. *IEEE Transactions on Robotics* **24**, 416–429 (2008).
- [29] Hogan, N. Impedance Control: An Approach to Manipulation. *1984 American Control Conference* 304–313 (IEEE, 1984).
- [30] Albu-Schäffer, A., Ott, C. & Hirzinger, G. A Unified Passivity-based Control Framework for Position, Torque and Impedance Control of Flexible Joint Robots. *The International Journal of Robotics Research* **26**, 23–39 (2007).
- [31] Yang, C., Ganesh, G., Haddadin, S., Parusel, S., Albu-Schaeffer, A. & Burdet, E. Human-Like Adaptation of Force and Impedance in Stable and Unstable Interactions. *IEEE Transactions on Robotics* **27**, 918–930 (2011).
- [32] Biagiotti, L., Liu, H., Hirzinger, G. & Melchiorri, C. Cartesian Impedance Control for Dexterous Manipulation. *Proceedings 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems* 3270–3275 (IEEE, 2003).
- [33] Pfanne, M., Chalon, M., Stulp, F., Ritter, H. & Albu-Schaffer, A. Object-Level Impedance Control for Dexterous In-Hand Manipulation. *IEEE Robotics and Automation Letters* **5**, 2987–2994 (2020).

- [34] Petit, F. & Albu-Schaffer, A. Cartesian Impedance Control for a Variable Stiffness Robot Arm. *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems* 4180–4186 (IEEE, 2011).
- [35] Kronander, K. & Billard, A. Stability Considerations for Variable Impedance Control. *IEEE Transactions on Robotics* **32**, 1298–1305 (2016).
- [36] Pratt, J., Chew, C., Torres, A., Dilworth, P. & Pratt, G. Virtual Model Control: An Intuitive Approach for Bipedal Locomotion. *The International Journal of Robotics Research* **20**, 129–143 (2001).
- [37] Hogan, N. On the Stability of Manipulators Performing Contact Tasks. *IEEE Journal on Robotics and Automation* **4**, 677–686 (1988).
- [38] Gao, X., Yao, K., Junge, K., Hughes, J. & Billard, A. A Detachable Crawling Robotic Hand. *Nature Communications* **17**, 428 (2026).
- [39] Junge, K. & Hughes, J. ADAPT-Teleop: Robotic Hand with Human Matched Embodiment Enables Dexterous Teleoperated Manipulation. *npj Robotics* **3**, 31 (2025).
- [40] Zhang, Y., Wang, H. & Hughes, J. Modular Anthropomorphic Hand Design via Multi-Parameter Finger Benchmarking and Selection. Preprint at <https://arxiv.org/abs/2606.11826> (2026).
- [41] Junge, K. & Hughes, J. Spatially Distributed Biomimetic Compliance Enables Robust Anthropomorphic Robotic Manipulation. *Communications Engineering* **4**, 76 (2025).
- [42] Hatanaka, T., Chopra, N. & Spong, M. W. Passivity-Based Control of Robots: Historical Perspective and Contemporary Issues. *2015 54th IEEE Conference on Decision and Control (CDC)* 2450–2452 (IEEE, 2015).

- [43] Larby, D. & Forni, F. A Generalized Approach to Impedance Control Design for Robotic Minimally Invasive Surgery. *IFAC-PapersOnLine* **56**, 8548–8555 (2023).
- [44] Chen, G., Guo, S., Hou, B. & Wang, J. Virtual Model Control for Quadruped Robots. *IEEE Access* **8**, 140736–140751 (2020).
- [45] Al-Shuka, H. F. N., Leonhardt, S., Zhu, W., Song, R., Ding, C. & Li, Y. Active Impedance Control of Bioinspired Motion Robotic Manipulators: An Overview. *Applied Bionics and Biomechanics* **2018**, 1–19 (2018).
- [46] Zhang, Y., Larby, D., Iida, F. & Forni, F. Virtual Model Control for Compliant Reaching under Uncertainties. *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 795–801 (IEEE, 2024).
- [47] Larby, D. & Forni, F. Optimal Virtual Model Control for Robotics: Design and Tuning of Passivity-Based Controllers. *IEEE Transactions on Robotics* **42**, 439–454 (2026).
- [48] Abu-Dakka, F. J. & Saveriano, M. Variable Impedance Control and Learning—A Review. *Frontiers in Robotics and AI* **7**, 590681 (2020).
- [49] Zaidi, L., Corrales, J. A., Bouzgarrou, B. C., Mezouar, Y. & Sabourin, L. Model-Based Strategy for Grasping 3D Deformable Objects Using a Multi-Fingered Robotic Hand. *Robotics and Autonomous Systems* **95**, 196–206 (2017).
- [50] Zeng, G. & Hemami, A. An Overview of Robot Force Control. *Robotica* **15**, 473–482 (1997).
- [51] Petit, F. & Albu-Schaffer, A. State Feedback Damping Control for a Multi DOF Variable Stiffness Robot Arm. *2011 IEEE International Conference on Robotics and Automation (ICRA)* 5561–5567 (IEEE, 2011).

- [52] Santello, M., Flanders, M. & Soechting, J. F. Postural Hand Synergies for Tool Use. *Journal of Neuroscience* **18**, 10105–10115 (1998).
- [53] Wimbock, T., Jahn, B. & Hirzinger, G. Synergy Level Impedance Control for Multifingered Hands. *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems* 973–979 (IEEE, 2011).
- [54] Larby, D. & Forni, F. A Passivity Preserving H-infinity Synthesis Technique for Robot Control. *2022 IEEE 61st Conference on Decision and Control (CDC)* 1416–1422 (IEEE, 2022).
- [55] Murray, R. M., Li, Z. & Sastry, S. S. *A Mathematical Introduction to Robotic Manipulation* (CRC Press, 2017).
- [56] Ferraguti, F., Secchi, C. & Fantuzzi, C. A Tank-Based Approach to Impedance Control with Variable Stiffness. *2013 IEEE International Conference on Robotics and Automation* 4948–4953 (IEEE, 2013).
- [57] Salisbury, J. Active Stiffness Control of a Manipulator in Cartesian Coordinates. *1980 19th IEEE Conference on Decision and Control* 95–100 (IEEE, 1980).
- [58] Chen, S. & Kao, I. Conservative Congruence Transformation for Joint and Cartesian Stiffness Matrices of Robotic Hands and Fingers. *The International Journal of Robotics Research* **19**, 835–847 (2000).
- [59] Wagner, C. R. & Emmanouil, E. Efficiency and Power Limits of Electrical and Tendon-Sheath Transmissions for Surgical Robotics. *Frontiers in Robotics and AI* **5**, 50 (2018).
- [60] Wahrburg, A., Bos, J., Listmann, K. D., Dai, F., Matthias, B. & Ding, H. Motor-Current-Based Estimation of Cartesian Contact Forces and Torques for Robotic Manipulators and

Its Application to Force Control. *IEEE Transactions on Automation Science and Engineering* **15**, 879–886 (2018).

- [61] Van Damme, M., Beyl, P., Vanderborght, B., Grosu, V., Van Ham, R., Vanderniepen, I. et al. Estimating Robot End-Effector Force from Noisy Actuator Torque Measurements. *2011 IEEE International Conference on Robotics and Automation* 1108–1113 (IEEE, 2011).
- [62] Nakajima, K. & Fischer, I. (eds.) *Reservoir Computing: Theory, Physical Implementations, and Applications* (Springer Singapore, 2021).
- [63] Márton, L. & Lantos, B. Control of Mechanical Systems with Stribeck Friction and Backlash. *Systems & Control Letters* **58**, 141–147 (2009).
- [64] Goodman, D. & Kelso, J. A. S. Exploring the Functional Significance of Physiological Tremor: A Biospectroscopic Approach. *Experimental Brain Research* **49**, 419–431 (1983).

Supplementary Material

We provide additional data and specifications that we use to support the statements in the main text.

The Supplementary Material is organized as follows:

- **A: System characterization**
- **B: Description of additional experiments**
- **C: Description of videos**
- **D: VMC control, theoretical formulation for the apparatus**
- **E: Full force of VMC Theorem (force+stiffness, I order) and CCT correction**
- **F: Optimization for online tuning of force and stiffness**

A: System characterization

This section contains the characterization, so:

1. **Finger kinematics**
2. **Finger friction compensation validation**
3. **Hand kinematics**

A.1: Finger kinematics

A.3: Finger friction compensation validation

A.2: Hand kinematics

B: Description of additional experiments

This section contains the description of additional experiments and characterizations performed to support the results in the main text, and is divided in the following sections:

1. **Softening on contact**
2. **Sigmoidal and polynomial stiffness sweep (switch and increasing)**
3. **Force prediction for various VMC configurations (task-space, etc)**

B.1: Softening on contact

B.2: Sigmoidal and polynomial stiffness sweep (switch and increasing)

B.3: Force prediction for various VMC configurations (task-space, etc)

C: Description of videos

D: VMC control, theoretical formulation for the apparatus

E: Full force of VMC Theorem (force+stiffness, I order) and CCT correction

F: Optimization for online tuning of force and stiffness